KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 1st Term Examination, 2018
Department of Computer Science and Engineering
CSE 4115
Computer and Network Security

TIME: 3 hours FULL MARKS: 210

N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) Through Shannon characteristics, discuss the characteristics of good cipher. Define the terms: 'confusion' and 'diffusion'. (10)
   b) Define known-plaintext attack. How it can be associated with RSA cryptosystem? (10)
   c) On same modulus using RSA, how multiple parties can encrypt data? (09)
   d) Show that "RSA cryptosystem is homomorphic". (06)

2. a) Discuss properties of digital signature. Explain ElGamal digital signature algorithm with an example. (12)
   b) Explain the working procedure of ElGamal cryptosystem specifying that it is based on discrete logarithm problem. (10)
   c) Make a comparison between the mechanism of AES and DES. Describe AES algorithm using a block diagram. (13)

3. a) Define cryptographic hash function. Discuss properties of it. (07)
   b) How and why you combine digital signature and public key cryptography? Explain. (08)
   c) Discuss the parameters that make a computer network vulnerable. (10)
   d) Define sensitive data. Discuss the factors that can make data sensitive. (10)

4. a) How documents can be signed using public key cryptography and one way hash function? Discuss. (07)
   b) Discuss 'stream cipher' and 'block cipher'. Explain a specific key exchange protocol to distribute a key. (13)
   c) How 'primitive element' relates with ElGamal? (08)
   d) What are the requirements of database security? Explain them. (07)

SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) Give an intuitive explanation of the working principle of Elliptic Curve Cryptography (ECC). Also explain how encryption and decryption work in ECC. (13)
   b) Explain ECC based digital signature with mathematical proof. (10)
   c) What are the reasons to prefer ECC over RSA? (05)
   d) What is tapjacking? Explain the prevention mechanism used against tapjacking in context of Android operating system. (07)

6. a) Write short notes on following attacks: (16)
      i. Page-in-the-Middle. ii. Program Download Substitution.
      iii. SQL Injection. iv. Man-in-the-Browser.
   b) Explain how attackers use User-in-the-Middle attack to bypass CAPTCHA protection. (06)
   c) What are web bugs? Explain how websites can be protected against unwanted change both from administrator's and users' perspective. (13)

7. a) With appropriate code and stack diagram explain how Buffer Overflow can be exploited to execute attacker's provided code. Also explain some common counter measures that are used against Buffer Overflow attack. (18)
   b) Explain the uses of Format String attack with appropriate example(s) and stack diagrams. (11)
   c) Explain Integer Overflow attack using suitable example(s). (06)

8. a) Write short notes on the followings: (14)
      i. Salami Attack. ii. Covert Channel.
      iii. Document Virus using Microsoft's Dynamic Data Exchange (DDE).
   b) Explain the problems associated with Access Control Directory. (06)
   c) Explain the problems associated with Segmentation and Paging. Also discuss how combining Paging with Segmentation provides better performance for protection in general purpose operating system. (15)